

Transformer Mechanics Worksheet — SET D

Question 1: Tokenization

(a) Split “INTERNATIONALIZATION” into three reasonable sub-word pieces.

Answer:

A reasonable split is:

INTER | NATIONAL | IZATION

Justification:

- **INTER** is a common prefix meaning *between* or *among*.
- **NATIONAL** is the main root of the word and relates to a nation or country.
- **IZATION** is a common suffix used to indicate the process of making or becoming something.

Modern Transformer models generally use **sub-word tokenization** methods such as **Byte Pair Encoding (BPE)**, **WordPiece**, or **SentencePiece**. Instead of treating a long word as one token, these methods divide it into smaller meaningful units that are already present in the vocabulary. This improves the model’s ability to process rare and complex words.

(b) Why does sub-word tokenization help a model handle this word?

Answer:

Sub-word tokenization allows the model to process words that it has never seen before by breaking them into familiar parts. Even if the complete word “INTERNATIONALIZATION” is not present in the training data, the model has likely learned the meanings of **INTER**, **NATIONAL**, and **IZATION**.

Mechanism:

1. The input sentence is first divided into tokens.
2. Rare or unknown words are split into smaller sub-word units.
3. Each sub-word token is matched with an entry in the model’s vocabulary.
4. These tokens are converted into embeddings and processed by the Transformer.

Advantages of Sub-word Tokenization:

- Reduces Out-of-Vocabulary (OOV) words.
- Keeps the vocabulary size manageable.
- Helps the model understand rare, long, and newly formed words.

Question 2: Embeddings

(a) Which pair would have higher cosine similarity, and why?

Answer:

The pair **King** and **Queen** would have a much higher cosine similarity than **King** and **Apple**.

This is because **King** and **Queen** belong to the same semantic category of royalty. They frequently appear in similar contexts and share many linguistic relationships. As a result, their embedding vectors are close together in the embedding space.

In contrast, **Apple** represents a fruit and has no semantic relationship with **King**. Therefore, the cosine similarity between **King** and **Apple** is much lower.

(b) What does an embedding represent, and how is it different from a one-hot vector?

Answer:

An **embedding** is a dense numerical vector that represents the meaning and context of a word. During training, embeddings are learned automatically so that words with similar meanings have similar vector representations.

For example:

- King $\rightarrow$ [0.45, -0.62, 0.71, ...]
- Queen $\rightarrow$ [0.48, -0.60, 0.74, ...]

Since these vectors are similar, the model understands that the words are semantically related.

A **one-hot vector** is a sparse binary vector that contains a single **1** and all remaining values as **0**. It identifies a word but does not capture any semantic relationship between different words.

Mechanism:

1. Each token generated by the tokenizer is assigned a unique ID.
2. The ID is mapped to a dense embedding vector using an embedding matrix.
3. These embedding vectors become the input to the Transformer.
4. During training, the embedding vectors are updated so that words used in similar contexts become closer in the vector space.

Difference between Embedding and One-Hot Vector

Embedding	One-Hot Vector
Dense numerical vector	Sparse binary vector
Represents semantic meaning	Represents only word identity
Learned during training	Fixed representation
Similar words have similar vectors	No relationship between words

Question 3: Self-Attention Worked Example

Sentence:

“The city councilmen refused the demonstrators a permit because they feared violence.”

(a) The word “they” refers to:

Answer: Councilmen

(b) Why is self-attention needed?

Answer:

Self-attention enables the Transformer to examine every word in the sentence while interpreting another word. Instead of relying only on word order or the nearest noun, it considers the complete context.

In this sentence, the phrase **“because they feared violence”** explains the reason for refusing the permit. The model correctly identifies that **“they”** refers to **city councilmen**, because it is more logical that the councilmen feared violence and therefore refused the permit. A simple word-order rule might incorrectly choose the nearer noun, **demonstrators**.

Mechanism of Self-Attention:

1. Every word is converted into Query (Q), Key (K), and Value (V) vectors.
2. The Query of **“they”** is compared with the Keys of all the words in the sentence.
3. Similarity scores are calculated using the dot product.
4. The scores are normalized using the Softmax function.
5. Higher attention weights are assigned to the most relevant words.
6. The corresponding Value vectors are combined to create the final contextual representation of **“they.”**

This mechanism enables the Transformer to understand long-distance relationships and contextual meaning more accurately than traditional sequence models.

Question 4: Query, Key, and Value

Answer:

The self-attention mechanism uses three vectors for every input word: **Query (Q)**, **Key (K)**, and **Value (V)**.

Query (Q)

The Query vector represents what the current word is searching for in the sentence. It acts like a request for relevant information.

Key (K)

The Key vector represents the characteristics or information each word provides. It is compared with the Query vector to determine relevance.

Value (V)

The Value vector contains the actual information that contributes to the final output when attention is assigned.

Mechanism of Query, Key, and Value

1. Every input embedding is multiplied by three learned weight matrices to generate the Query, Key, and Value vectors.
2. The Query vector is compared with the Key vectors of all words using the dot product.
3. The similarity scores are scaled by dividing them by $\sqrt{d_k}$.
4. The Softmax function converts these scores into attention weights.
5. The attention weights are multiplied with the corresponding Value vectors.
6. The weighted Value vectors are summed to produce the final context-aware representation of the word.
7. This representation is passed to the next Transformer layer for further processing.

The attention computation is represented as:

$$[(Q, K, V) = ()V]$$

where:

- **Q** = Query matrix
- **K** = Key matrix
- **V** = Value matrix
- **d_k** = Dimension of the Key vectors

This mechanism allows the Transformer to focus on the most relevant words, capture long-range dependencies, resolve references, and understand the context of a sentence effectively.